

Determine whether each of the following functions are injective (one-to-one), surjective (onto), both or neither following the definitions from One-to-One and Onto Intro.pdf or

Section 12.2:

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 2x + 3$$

$$g : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 3$$

$$h : \mathbb{R} \rightarrow \mathbb{R}, h(x) = x^2$$

$f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 2x + 3$
 Injective (One-to-One)
 $x, y \in f(x), x \neq y \Rightarrow f(x) \neq f(y)$
 Suppose, contrapositively, that:
 $x = y \Rightarrow f(x) = f(y)$
 $2x + 3 = 2y + 3$
 $2x = 2y$
 $x = y$
 Since it is true contrapositively,
 $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is injective.
 Surjective (Onto)
 $\forall y \in \mathbb{R}, \exists x \in \mathbb{R}, s. t. y = 2x + 3$
 Since the range is declared as $\mathbb{R}$ and the domain is $\mathbb{R}$ we can see there is an infinite amount of x for y .
 Thus, it is surjective.
 This function is both

$g : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 3$
 Injective (One-to-One)
 $x, y \in f(x), x \neq y \Rightarrow f(x) \neq f(y)$
 Suppose, contrapositively, that:
 $x = y \Rightarrow f(x) = f(y)$
 $2x + 3 = 2y + 3$
 $2x = 2y$
 $x = y$
 Since it is true contrapositively,
 $f : \mathbb{R} \rightarrow \mathbb{R}$ is injective.
 Surjective (Onto)
 $\forall y \in \mathbb{R}, \exists x \in \mathbb{R}, s. t. y = 2x + 3$
 Since the range is declared as $\mathbb{R}$ and the domain is $\mathbb{R}$ we can see there is an infinite amount of x for y .
 Thus, it is surjective.
 This function is both

$h : \mathbb{R} \rightarrow \mathbb{R}, h(x) = x^2$
 Injective (One-to-One)
 $x, y \in f(x), x \neq y \Rightarrow f(x) \neq f(y)$
 Suppose, contrapositively, that:
 $x = y \Rightarrow f(x) = f(y)$
 $x^2 = y^2$
 Since it is true contrapositively,
 $f : \mathbb{R} \rightarrow \mathbb{R}$ is injective.
 Surjective (Onto)
 $\forall y \in \mathbb{R}, \exists x \in \mathbb{R}, s. t. y = x^2$
 Since the range is declared as $\mathbb{R}$ but would end up as $\mathbb{N}$ and the domain is $\mathbb{R}$ we can see there is an infinite amount of x for y .
 Thus, it is surjective.
 This function is both